

Day 9 is complete. Next in the fixed schedule is Day 10: ICMP, ping, trace Day 9 is complete. Next in the fixed schedule is Day 10: ICMP, ping, traceroute, and how Linux discovers where a network path is failing. route, and how Linux discovers where a network path is failing. -

Day 10 — ICMP, ping, traceroute, route & Network Troubleshooting

Today we connect **IP + Routing + ICMP** and learn how Linux can determine **where network communication is failing**.

This is a very important interview topic because interviewers often ask:

"What happens when you run ping?"

or

"How does traceroute find each router?"

or

"How would you troubleshoot a network problem in Linux?"

Day 10 Goals

By the end of today, you should understand:

```
ICMP
↓
ping
↓
ICMP Echo Request / Echo Reply
↓
TTL
↓
traceroute
↓
Intermediate routers
↓
Linux network troubleshooting
```

And these commands:

```
ping
traceroute
ip route
ip route get
ip addr
ip link
ss
ip neigh
```

1. What is ICMP?

ICMP = Internet Control Message Protocol

ICMP operates alongside IP at the network layer and is used for:

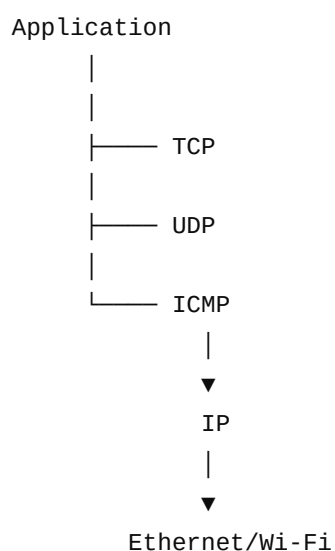
- Error reporting
- Diagnostics
- Network reachability
- Control/information messages

Important:

ICMP is **not TCP and not UDP**.

It is carried directly inside IP packets.

Conceptually:



2. TCP vs UDP vs ICMP

You already know:

TCP
↓
Transport protocol
↓
Ports

UDP
↓
Transport protocol
↓
Ports

ICMP is different:

ICMP
↓
Network-layer control/diagnostic protocol
↓
No TCP/UDP port

This is a common interview question.

? Does ICMP use port 80?

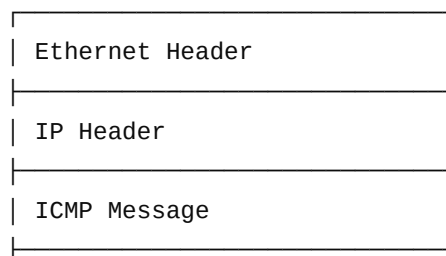
No.

ICMP doesn't use TCP/UDP ports.

3. Where Does ICMP Exist?

Think of the packet structure:

Ethernet Frame



So:

Ethernet



IP



ICMP

Not:

IP



TCP



ICMP

4. Why Was ICMP Created?

Imagine your PC sends:

Packet → Router → Router → Router → Server

Something could go wrong:

Destination unreachable



TTL expired



Network unreachable



Host unreachable

The sender needs a way to know about certain network-layer problems.

ICMP provides messages for this.

5. Common ICMP Messages

You should know these:

ICMP Message	Meaning
Echo Request	"Are you reachable?"

ICMP Message	Meaning
Echo Reply	"Yes, I'm reachable."
Destination Unreachable	Destination/network cannot be reached
Time Exceeded	TTL reached zero
Redirect	Routing-related notification

For today's interview preparation, focus heavily on:

Echo Request
Echo Reply
Time Exceeded
Destination Unreachable

6. What is ping?

ping is a network diagnostic utility that commonly uses:

ICMP Echo Request

and:

ICMP Echo Reply

Example:

ping 8.8.8.8

Conceptually:

```
PC-A
|
| ICMP Echo Request
▼
Server
|
| ICMP Echo Reply
▼
PC-A
```

7. Ping — Step-by-Step

Suppose:

```
PC-A  
192.168.1.10
```

runs:

```
ping 192.168.1.20
```

Step 1 — Application starts ping

```
ping  
|  
▼  
ICMP Echo Request
```

Step 2 — IP packet created

```
Source IP      = 192.168.1.10  
Destination IP = 192.168.1.20
```

Step 3 — Routing lookup

Linux checks:

```
Is 192.168.1.20  
on the local subnet?
```

Suppose:

```
192.168.1.0/24
```

Yes.

Step 4 — ARP

```
Who has 192.168.1.20?
```

PC-B responds with its MAC.

Step 5 — Ethernet

PC-A MAC → PC-B MAC

Step 6 — PC-B receives

PC-B sees:

ICMP Echo Request

Step 7 — PC-B responds

ICMP Echo Reply

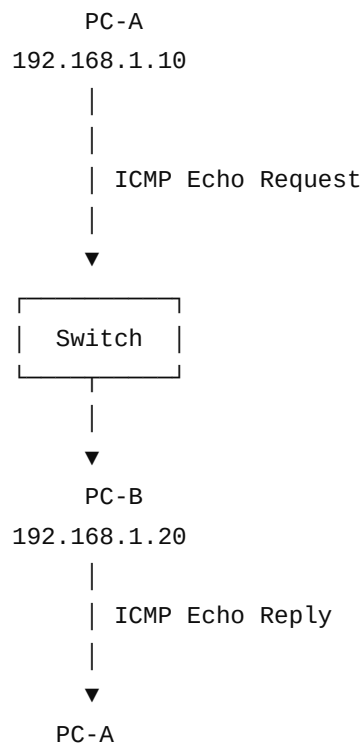
Step 8 — PC-A receives reply

Echo Reply

↓

ping calculates RTT

8. Ping Visualization



Notice:

```
ICMP
↓
IP
↓
Ethernet
↓
NIC
```

No TCP connection is created.

No UDP port is involved.

9. What Does Ping Actually Measure?

Example:

```
64 bytes from 8.8.8.8:
time=20 ms
```

The 20 ms is approximately:

```
PC-A
|
| Echo Request
▼
Destination
|
| Echo Reply
▼
PC-A
```

This is **round-trip time (RTT)**.

```
RTT = request time + reply time
```

10. Ping Does NOT Mean "Internet Is Perfect"

Suppose:

```
ping 8.8.8.8
```

works.

That tells us that:

```
ICMP Echo Request
  ↓
Destination
  ↓
ICMP Echo Reply
```

worked.

But it does **not** prove:

- DNS works
- HTTP works
- HTTPS works
- Your application works
- TCP port 443 is reachable

For example:

```
ping works
  |
  └─ ICMP works
  |
  └─ TCP 443 might still fail
```

A firewall may allow ICMP but block TCP/443, or vice versa.

11. Why Does `ping google.com` Involve DNS?

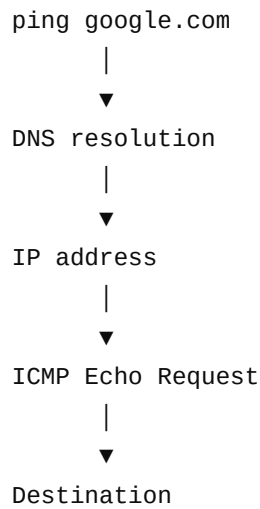
When you run:

```
ping google.com
```

`ping` first needs to resolve:

```
google.com
  ↓
IP address
```

So the overall process can involve:



Therefore:

If `ping 8.8.8.8` works but `ping google.com` fails, DNS may be the problem.

This is a very useful troubleshooting clue.

12. What is TTL?

We introduced TTL in Day 9.

TTL = Time To Live

IPv4 has a TTL field.

Example:

TTL = 64

When a router forwards the packet:

```
graph TD; A[64] --> B[63]; B --> C[62]; C --> D[61];
```

A vertical sequence of numbers showing the TTL value decreasing by 1 at each step. It starts with 64, followed by a downward arrow to 63, another arrow to 62, and a final arrow to 61.

Each router decreases it.

13. Why Do We Need TTL?

Imagine a routing loop:

```
Router A
  ↓
Router B
  ↓
Router C
  ↓
Router A
  ↓
Router B
  ↓
...
```

Without TTL, the packet could circulate indefinitely.

TTL prevents that.

Eventually:

```
TTL → 0
```

The router discards the packet.

14. What Does the Router Send Back?

When TTL expires, the router can send:

```
ICMP Time Exceeded
```

back toward the sender.

This is the key mechanism behind **traceroute**.

15. What is traceroute?

traceroute attempts to show:

Which routers are present along the path to a destination?

Example:

```
traceroute 8.8.8.8
```

Conceptually:

```
PC
|
▼
Router 1
|
▼
Router 2
|
▼
Router 3
|
▼
8.8.8.8
```

Traceroute tries to discover:

```
Hop 1
Hop 2
Hop 3
...
Destination
```

16. How Does Traceroute Discover Routers?

This is one of the best interview questions.

Suppose traceroute sends a packet with:

TTL = 1

Router 1

Packet arrives:

TTL = 1

Router decrements:

1 → 0

Packet is discarded.

Router sends:

ICMP Time Exceeded

back to your PC.

Therefore your PC learns:

Hop 1 = Router 1

17. TTL = 2

Traceroute sends another packet:

TTL = 2

Router 1:

2 → 1

Forwards it.

Router 2:

1 → 0

Router 2 discards it.

Router 2 sends:

ICMP Time Exceeded

Now:

Hop 1 = Router 1

Hop 2 = Router 2

18. TTL = 3

TTL = 3

Router 1:

3 → 2

Router 2:

2 → 1

Router 3:

1 → 0

Router 3 responds:

ICMP Time Exceeded

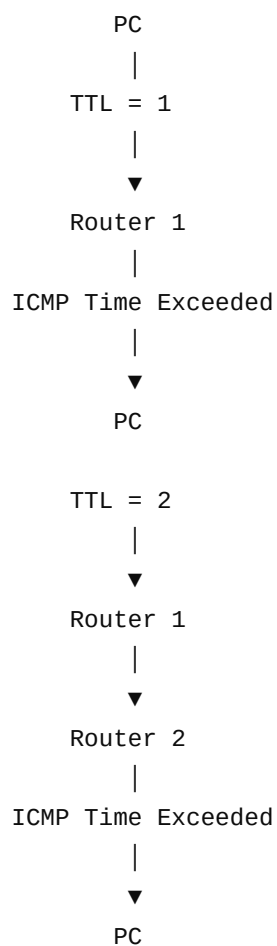
Now traceroute knows:

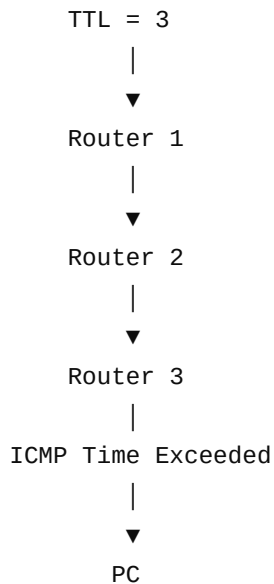
Hop 1 → Router 1

Hop 2 → Router 2

Hop 3 → Router 3

19. Traceroute Visualization





20. Eventually TTL Reaches the Destination

Suppose:

Destination = 8.8.8.8

Eventually traceroute sends a packet whose TTL is large enough to reach the destination.

The destination responds according to the traceroute method being used.

Traceroute then knows:

Destination reached

21. Linux traceroute vs tracepath

On Linux you may see:

traceroute 8.8.8.8

and:

tracepath 8.8.8.8

tracpath is often available on Linux systems and can be useful for path and MTU diagnostics.

There are also different traceroute probe methods depending on implementation/options.

For interview purposes, remember the fundamental idea:

```
Traceroute
↓
Manipulates TTL
↓
Routers decrement TTL
↓
TTL expires
↓
ICMP Time Exceeded
↓
Sender learns intermediate hop
```

22. Why Do Some Hops Show * * * ?

You might see:

```
1  192.168.1.1
2  10.x.x.x
3  * * *
4  ...
```

* generally means traceroute did not receive a response to that probe within the expected time.

Possible reasons include:

```
Firewall
ICMP filtering
Rate limiting
Router configuration
Packet loss
Asymmetric path
```

Important:

* * * does **not automatically mean the network is broken.**

The router may simply not respond to traceroute probes.

23. What is route ?

Historically Linux systems used:

```
route -n
```

to display routing information.

Today, the preferred modern command is:

```
ip route
```

So for your interview:

```
Old:  
route
```

```
Modern:  
ip route
```

You should be comfortable with `ip route` .

24. ip route

Example:

```
ip route
```

Output might look like:

```
default via 192.168.1.1 dev eth0  
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10
```

Interpretation:

```
default  
↓  
Gateway = 192.168.1.1
```



and:

```
192.168.1.0/24
  ↓
Directly connected
  ↓
eth0
```

25. ip route get

This is one of my favorite Linux networking commands for interviews.

Run:

```
ip route get 8.8.8.8
```

It asks Linux:

"If I want to reach this destination, what route would you use?"

Example:

```
8.8.8.8 via 192.168.1.1 dev eth0
```

Meaning:

```
Destination
8.8.8.8
  ↓
Next Hop
192.168.1.1
  ↓
Interface
eth0
```

This is extremely useful when debugging routing.

26. ip addr

To inspect IP addresses:

```
ip addr
```

or:

```
ip a
```

You might see:

```
eth0:
    inet 192.168.1.10/24
```

This tells you:

```
Interface = eth0
IP         = 192.168.1.10
Prefix     = /24
```

27. ip link

To check network interfaces:

```
ip link
```

Conceptually:

```
eth0 → UP
lo    → UP
```

You can determine whether an interface is:

```
UP
DOWN
```

28. ip neigh

Remember ARP from Day 7.

Linux maintains neighbor information.

Use:

```
ip neigh
```

Example:

```
192.168.1.1 dev eth0  
lladdr aa:bb:cc:dd:ee:ff  
REACHABLE
```

This tells you:

```
IP:  
192.168.1.1  
  
MAC:  
aa:bb:cc:dd:ee:ff
```

This is extremely useful when diagnosing:

```
ARP problems  
Gateway problems  
Local LAN connectivity
```

29. ss

To inspect sockets:

```
ss -tuln
```

Meaning:

```
-t → TCP  
-u → UDP  
-l → listening  
-n → don't resolve names
```

Example:

```
LISTEN
0.0.0.0:22
```

```
LISTEN
0.0.0.0:8080
```

This tells you that applications are listening on:

```
22
8080
```

30. Network Troubleshooting Flow

Now let's create an interview-ready troubleshooting procedure.

Suppose:

"My application cannot connect to a server."

Don't randomly run commands.

Follow layers.

```
Application
  |
  ▼
Socket
  |
  ▼
TCP/UDP
  |
  ▼
IP
  |
  ▼
Routing
  |
  ▼
ARP/Neighbor
  |
  ▼
NIC
  |
  ▼
Network
```

31. Step 1 — Check Interface

```
ip link
```

Ask:

```
Is eth0/ens... UP?
```

If:

```
DOWN
```

you have a basic interface problem.

32. Step 2 — Check IP Address

```
ip addr
```

Check:

```
IP address  
Subnet prefix  
Interface
```

Example:

```
192.168.1.10/24
```

33. Step 3 — Check Routing

```
ip route
```

Check:

```
Default gateway  
Destination route
```



Then:

```
ip route get 8.8.8.8
```

34. Step 4 — Check Neighbor/ARP

```
ip neigh
```

Look for:

```
Gateway IP  
MAC address  
REACHABLE/STALE/FAILED
```

If gateway resolution is failing:

```
PC  
↓  
ARP  
↓  
Gateway MAC
```

may be the problem.

35. Step 5 — Ping Gateway

Suppose gateway:

```
192.168.1.1
```

Run:

```
ping 192.168.1.1
```

If this fails:

PC → Gateway

is already broken.

Potential causes:

NIC
Cable/Wi-Fi
VLAN
ARP
Gateway
Local firewall

36. Step 6 — Ping Remote IP

If gateway works:

ping 8.8.8.8

If:

gateway works
remote IP fails

the problem may be farther along the path.

PC
↓
Gateway ✓
↓
Internet ✗

37. Step 7 — Test DNS

If:

ping 8.8.8.8

works but:


```
ping google.com
```

fails:

Think:

```
DNS problem
```

Check DNS configuration with tools such as:

```
resolvectl status
```

on systems using `systemd-resolved`.

38. Step 8 — Check TCP Port

Suppose:

```
Server = 192.168.2.20  
Port = 8080
```

Ping may work but the application still fails.

Why?

Because:

```
ICMP ≠ TCP
```

You can test TCP connectivity with tools such as:

```
nc -vz 192.168.2.20 8080
```

Conceptually:

```
ICMP works  
  ↓  
But TCP 8080?  
  ↓  
Test separately
```

39. Complete Troubleshooting Ladder

Memorize this:

```
1. Interface
  ↓
  ip link

2. IP address
  ↓
  ip addr

3. Routing
  ↓
  ip route

4. Neighbor / ARP
  ↓
  ip neigh

5. Gateway
  ↓
  ping gateway

6. Remote IP
  ↓
  ping remote-IP

7. DNS
  ↓
  resolve hostname

8. TCP port
  ↓
  nc / application test

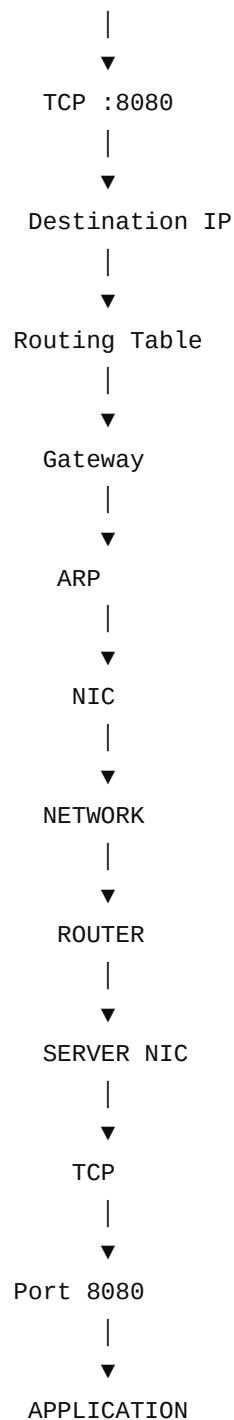
9. Application
  ↓
  logs / configuration
```

This is a very good interview answer.

40. Full PC → Server Troubleshooting Diagram

```
APPLICATION
|
connect()
```





If something fails, identify **which layer**.

41. Important Interview Scenario

Interviewer:

"Ping is working but my application cannot connect. What will you check?"

Good answer:

Ping works

↓

Basic IP connectivity exists

↓

But ICMP working doesn't guarantee TCP application connectivity

↓

Check:

- destination IP
- destination port
- server listening socket
- firewall
- routing
- application status

On Linux:

```
ss -ltn
ip route
ip route get <destination>
nc -vz <destination> <port>
```

42. Another Interview Scenario

Interviewer:

" ping 8.8.8.8 works, but ping google.com doesn't. What could be wrong?"

Answer:

```
8.8.8.8
↓
ICMP works

google.com
↓
DNS resolution required
↓
DNS may be failing
```

So check:

```
resolvectl status
```

and DNS configuration/connectivity.

43. Another Scenario

Interviewer:

"I can ping the gateway but cannot reach the Internet."

Think:

```
PC
|
├─ NIC ✓
|
├─ IP ✓
|
├─ ARP ✓
|
├─ Gateway ✓
|
└─ Beyond gateway x
```

Possible causes:

```
Routing problem
Gateway configuration
WAN/ISP problem
Firewall
NAT
Upstream routing
```

You can investigate with:

```
ip route
tracert <destination>
```

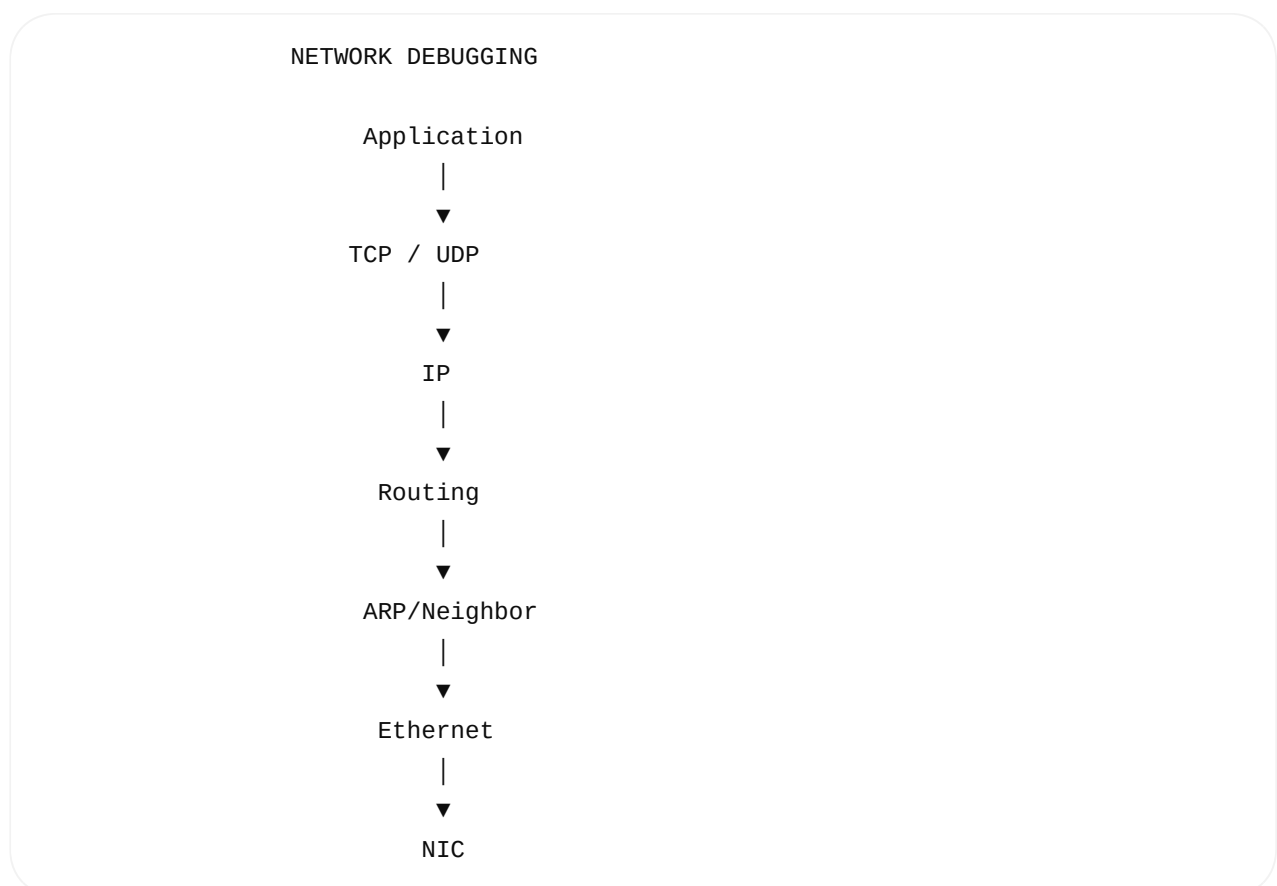
44. ping vs traceroute

Tool	Main purpose
ping	Test reachability + RTT
tracert	Discover path/hops
ip route	Inspect routing table

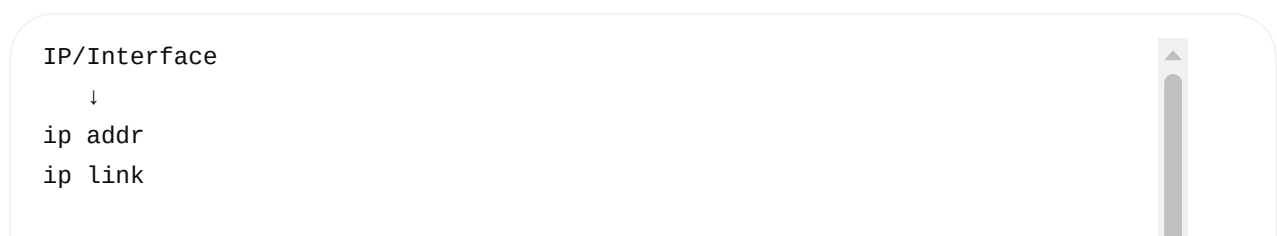
Tool	Main purpose
<code>ip route get</code>	Determine route to destination
<code>ip neigh</code>	Inspect IP ↔ MAC neighbor mapping
<code>ip addr</code>	Inspect IP configuration
<code>ip link</code>	Inspect interface state
<code>ss</code>	Inspect sockets
<code>nc</code>	Test TCP/UDP connectivity

★ Day 10 — The Most Important Mental Model

Remember this:



Diagnostic tools map onto these areas:



Routing



ip route

ip route get

ARP



ip neigh

Reachability



ping

Path



traceroute / tracepath

Sockets



ss

Port connectivity



nc

Day 10 Interview Questions

Try answering these without looking back:

1. What is ICMP?
2. Does ICMP use TCP or UDP?
3. Does ICMP use port numbers?
4. What is an ICMP Echo Request?
5. What is an ICMP Echo Reply?
6. How does `ping` work?
7. What does ping's RTT represent?
8. Does successful ping guarantee TCP connectivity?
9. What is TTL?
10. Why does IPv4 have TTL?
11. What happens when TTL reaches zero?
12. What is ICMP Time Exceeded?
13. How does `traceroute` discover routers?
14. Why does `traceroute` use different TTL values?

15. What does * * * mean in traceroute?
 16. What is the modern Linux replacement for route ?
 17. What does ip route show?
 18. What does ip route get <IP> do?
 19. What does ip neigh show?
 20. How would you troubleshoot an application that cannot connect to a server?
-

★ Day 10 Must-Memorize

```
ping
↓
ICMP Echo Request
↓
IP
↓
Network
↓
ICMP Echo Reply
↓
RTT
```

```
traceroute
↓
TTL = 1
↓
Router 1
↓
ICMP Time Exceeded

TTL = 2
↓
Router 2
↓
ICMP Time Exceeded

TTL = 3
↓
Router 3
↓
ICMP Time Exceeded

...
↓
Destination
```


And the Linux troubleshooting sequence:

```
ip link
↓
ip addr
↓
ip route
↓
ip route get
↓
ip neigh
↓
ping gateway
↓
ping remote IP
↓
DNS test
↓
TCP port test
↓
Application
```

Day 10 is now complete. Next in the fixed schedule is Day 11 — DNS in depth:

hostname → DNS resolver → recursive DNS → root → TLD → authoritative server → IP address → application connection.